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A detailed palaeomagnetic study of the oldest (≈15 Myr) lava sequences in Northwest Iceland

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SUMMARY

We have determined the directions of primary remanence in over 400 lava flows in the northwestern peninsula of Iceland (near 66.1°N, 23.3°W). These lavas were sampled in 16 profiles spread across 75 km, at the stratigraphic level of the oldest (approximately 15 Myr in age) of several distinctive lignite-bearing sediments in the peninsula. We find that the pattern of polarity reversals in the lava pile is broadly similar in these profiles but it is generally not possible to trace short polarity zones and excursions over more than a few kilometres laterally. Our results, which also include measurements on the lignite sediments at three sites, are consistent with previous suggestions that these sediment beds represent a time gap of the order of 0.2 Myr between the lavas above and below. The mean virtual geomagnetic pole of this collection is 'far-sided' by approximately 7°, and the scatter of individual poles is greater than that in palaeomagnetic surveys on younger lava series in Iceland.

Key words: Cenozoic, Iceland, igneous rock, palaeomagnetism, sediments.

1 INTRODUCTION

1.1 General

Iceland owes its presence to a combination of crustal generation by sea floor spreading and the activity of a mantle plume. Outside of the regions generated during the last couple of Myr under glacial conditions, the exposed formations consist primarily of subaerially erupted lava flow sequences. The lavas originate partly in fissure eruptions and partly within central volcano complexes. Judging from the results of extensive radiometric dating in different areas of Iceland, any two adjacent Tertiary lava flows may be expected to be 5–10 kyr apart in time on average (see table 1 of Kristjansson & McDougall 1982), and the average thickness of a lava flow is of the order of 10 m (e.g. Walker 1959; Kristjansson & Johannesson 1996). The lavas can usually be distinguished from each other as their topmost parts tend to be scoriaceous. Interbasaltic red-brown volcanogenic sediments less than 0.5 m thick are also generally present.

Research on the remanent magnetization of Icelandic lavas was initiated on a small scale around 1930. Since 1950, this research has led to several important discoveries regarding the history of the geomagnetic field, including the first records of 'intermediate' directions at polarity zone boundaries, *cf.* the review by Kristjansson (1993a). The lava pile of Iceland presents virtually unlimited opportunities for palaeomagnetic research. Areas may be chosen where the lavas have suffered a minimum of thermal and tectonic disturbance, with good accessible exposures of stratigraphically overlapping profiles, each containing up to 50 or more lavas. In many areas of the country no stratigraphic mapping or palaeomagnetic studies have yet been carried out, and only preliminary work has been done in many other areas. The usefulness of past and future palaeomagnetic results from the lava pile may be greatly augmented by improved techniques of radiometric age determination, e.g. in accurate dating of geomagnetic reversals at favourable sites.

To this date, palaeomagnetic studies in Iceland have mainly been carried out for two purposes.

(i) In local stratigraphic correlations. In stratigraphic work, geologists have partly relied on tracing sediment beds, as well as groups of several lavas having abundant feldspar phenocrysts or other easily recognizable characteristics (Walker 1959). In East Iceland some of these lava groups may extend over tens of kilometres but others are less distinct, interfingering with the tholeiites which are the most common lava type. Chemical analyses have so far hardly been tested at all in stratigraphy in Iceland. A zone of constant palaeomagnetic polarity in the lava pile typically consists of 10–25 lava units (Kristjansson & McDougall 1982) and such zones therefore provide a useful supplement to the geologist's mapping methods. Polarities can in many cases be measured on hand samples with a magnetometer in the field, but this method is not recommended due to the frequent presence of viscous remanence (VRM) and other disturbing factors such as induced magnetization and shape effects.

(ii) As a means of inferring the behaviour of the palaeomagnetic field. This work has largely concentrated on various overall properties of the field. The time interval between flows is variable

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and may reach tens of thousands of years. Hence, detailed studies of pole paths in the secular variation or in individual reversals and excursions are only possible in locations where the build-up of the lava pile has been unusually rapid. Few such pole paths have been described from Iceland (see Kristjansson 1993a; Kristjansson & Sigurgeirsson 1993).

1.2 Northwest peninsula: previous work

The Northwest peninsula of Iceland (Fig. 1) is a remnant of a larger plateau of lavas, built up more or less continuously during the period from approximately 15 Myr ago to approximately 8 Myr ago (McDougall *et al.* 1984; Hardarson *et al.* 1997). The lava series in the peninsula reach further back in time than any rocks dated elsewhere in Iceland, as the oldest rocks in East Iceland are less than 14 Myr old (Watkins & Walker 1977; unpublished data quoted by Hardarson *et al.* 1997). Several central-volcano complexes are known in the peninsula but their hydrothermal and tectonic effects on the exposed lava pile are not as pronounced as in East Iceland, partly due to much less erosion and uplift in the northwest. Thus, in some of the fjords of East Iceland the mesolite zone of alteration reaches an altitude of 500 m or more (Watkins & Walker 1977, figs 3a and b) whereas in the area considered here, mesolite alteration is not seen above sea level (McDougall *et al.* 1984, fig. 3).

A feature of the geology of the NW peninsula which attracted the attention of scientists already in the 18th century, was the presence of interbasaltic lignites and leaf impressions at a large number of localities. The sediments of these sites are generally thick, up to tens of metres, whereas the lignites themselves occupy less than 1 m. They were for a long while (see e.g. Kristjansson 1993b) thought to belong to a single horizon between an Eocene lava sequence and a much younger (Miocene) one. It is now clear that in the western part of the peninsula there are at least four such horizons. By interpolating between K-Ar age determinations in the lava pile by McDougall *et al.* (1984, fig. 3) and Hardarson *et al.* (1997), and making use of stratigraphic correlations by Kristjansson & Johannesson (1996), we estimate their ages to be around 12, 12.5, 13 and 15 Myr, respectively. However, the sediments are often obscured by soil or scree cover, and many outcrops are too steep to be accessible. The sediments do not appear in some well-exposed profiles where they might be expected. Only one modern geological study (Roaldset 1983) has been carried out on the oldest lignite sediments. This study dealt with their alteration mineralogy, but little research on the sedimentology of these lignite-bearing beds or the environment of their formation, has been published since the 19th century.

Another unusual feature of the lava pile in the area of Fig. 1 is the occurrence of lava flows containing unusually large feldspar phenocrysts, up to 1 cm or more in size. A recent discovery is the presence of ankaramites in the upper parts of some of our profiles.

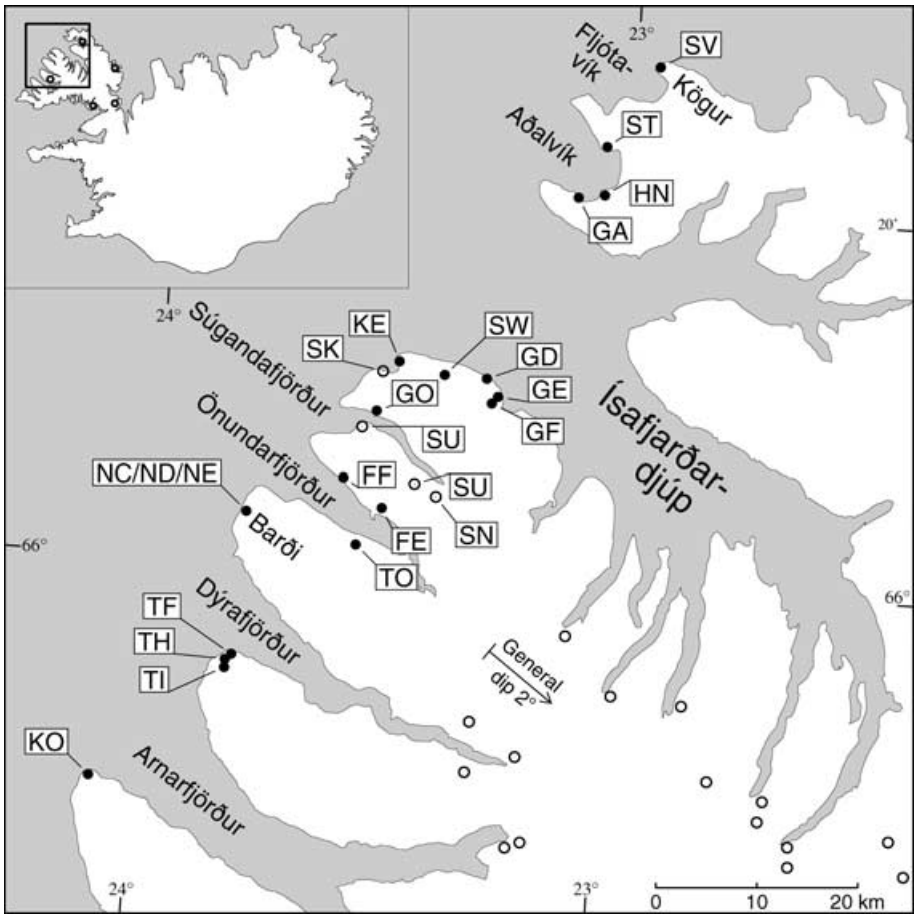


Figure 1. Northwest Iceland. The locations of sampling profiles in the present survey are shown as black dots, each with a two-letter code name. Those profiles in the survey of McDougall *et al.* (1984), which are included in Figs 3 and 4, are shown as circles with code names. The location of various other profiles in that survey and in the survey of Kristjansson & Johannesson (1996) are shown as unmarked circles. The small inlet Skalavik lies between profiles SK and KE, and the inlet Bolungarvik is that east of profiles GE and GF. In the inset, the approximate positions of the main volcanic centres of the NW peninsula are indicated.

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Kristjansson *et al.* (1975) and Kristjansson & Johannesson (1996) have reported that both of these lava types also occur sporadically around the stratigraphic level of the younger lignite-sediment beds in the NW peninsula.

The first palaeomagnetic studies in the NW peninsula were made by Kristjansson (1968) on 60 lava flows in the region of Bolungarvik–Önundarfjördur (see Fig. 1) and by Kristjansson *et al.* (1975) at one profile of 44 lava flows in Arnarfjördur (just south of Fig. 1). These authors reported that the lavas are generally good material for palaeomagnetic work, following removal of viscous magnetization by routine alternating field (AF) treatment at 15 or 20 mT peak fields.

A major palaeomagnetic collection in the NW peninsula was carried out by N.D. Watkins and co-workers, in a study also including radiometric dating (McDougall *et al.* 1984). About 500 lavas were sampled in 1975 in 10 mapped profiles, making up a composite section through the western part of the peninsula. Additionally, several hundred younger lavas were sampled in profiles on the south and east coasts of the peninsula in 1975–77. The oldest lignite sediments mentioned above are present in two of the profiles, named SK and SU (Fig. 1). However, the former profile was not sampled all the way up to the sediments. In the latter, exposures around the sediments are poor, so that McDougall *et al.* added a short profile SN approximately 2 km to the south of the segment where they were expected. The lava series in SN appears to be wedging out towards SU.

Another palaeomagnetic study of a composite section in the peninsula was carried out by Kristjansson & Johannesson (1996). This study, which includes 307 sampled flows in the tributary fjords south of the Isafjardardjup fjord (Fig. 1), begins at a stratigraphic level well above the oldest lignite sediments. Kristjansson & Johannesson (1996) correlated the polarity zone pattern found along Isafjardardjup with that of McDougall *et al.* (1984), on the assumption that a sediment bed near the top of their section represents the lignite sediments (of 12 Myr age) near the top of the section of McDougall *et al.* This correlation is yet to be tested by radiometric dating in the Isafjardardjup profiles.

Hardarson *et al.* (1997) attempted to use polarity zones for detailed correlations between three profiles along the strike of the lignite sediments: KE approximately 2 km northeast of the profile SK mentioned above (see Fig. 1), 'SU' on the south side of Sugandafjördur and TF south of Dyrafjördur. However, the 'SU' of Hardarson *et al.* (1997) is a composite, not identical to the SU profile studied by McDougall *et al.* (1984): 'SU' consists of flows SU 1–4 followed by polarity measurements in the field on ten flows at a nearby locality and then data from McDougall *et al.*'s flows SN 2–15 renamed 'SU' 18–30. Furthermore, all polarities reported by Hardarson *et al.* (1997) from profiles KE and TF were obtained in the field. Their polarity results and correlations will be revised considerably below, on the basis of laboratory magnetic measurements.

2 THE PRESENT STUDY

2.1 Geological considerations

The nature of the oldest lignite-bearing sediments in Iceland has long been debated, and much research is still needed to resolve this matter. Among questions remaining to be answered satisfactorily are the following.

(1) Are there measurable age differences between the lavas occurring immediately below the sediments at different sites? If so, do the lava ages change laterally in some systematic way? The same questions may also be asked for the lavas immediately above the sediments.

(2) Did regional tilting take place in the period between the deposition of these lavas and those above the sediments?

(3) Was there much erosion (even down to sea level) during the hiatus in lava emplacement? This might be reflected, for example, in an abrupt change in lava alteration state and dyke density at the level of the sediments.

(4) Did a systematic change in lava chemistry, rate of build-up or other characteristics take place at the hiatus?

(5) How long did it take to form the sediments which now remain in place? In the case where erosion stopped before the sediments were deposited, is there evidence for chemical weathering of the uppermost underlying lavas during that period?

At the moment the available radiometric ages (McDougall *et al.* 1984; Hardarson *et al.* 1997; M.S. Pringle & B.S. Hardarson, preliminary data) relevant to questions 1–3 have an uncertainty of hundreds of kyr. As to question 2, accurate measurements are lacking. Most tilt estimates so far published vary from 0° to 3° in a general southeasterly direction. They may to some extent reflect uncertainties in measurement as well as local variability. The dip begins to increase south of Dyrafjördur and reaches 6° at the south coast of the peninsula. The high values of 5° to the NW below the sediments and 5° to the SE above them, given by Hardarson *et al.* (1997, p. 183) do not apply to the profiles of this study. The general regional tilt (for the purpose of correcting the observed palaeomagnetic directions) is here assumed to be 2° to the SE at sea level, in some cases decreasing upwards. According to Hardarson *et al.* (1997) the answer to question 4 is affirmative in the three profiles they studied: indications of a change in the lava source are seen from both Zr/Nb ratios and isotopic data. These authors suggest that the change in chemistry is connected with rift relocation processes.

In this paper we attempt to throw light on some of the questions above, using results from new detailed laboratory measurements on the natural remanence of over 400 lava flows and three sites in the sedimentary interbeds, along with measurements on some 80 flows sampled in 1975 (McDougall *et al.* 1984). The work by Hardarson *et al.* (1997) on the geochemistry and isotope ratios in the lava pile in this area is being continued and will be reported elsewhere.

2.2 Purpose of the palaeomagnetic part of the present mapping project

According to Bodvarsson & Walker (1964) the lateral extent of a lava flow in Iceland is of the order of 1000 times its greatest thickness, i.e. on average 10 km. In the large stratigraphic projects carried out in the Icelandic lava pile in the 1960s and 1970s (e.g. Watkins & Walker 1977; McDougall *et al.* 1984), the distance between adjacent profiles sampled for palaeomagnetic laboratory studies was variable but often reached 10 or even 20 km (although some field measurements of lava polarity were carried out in between the sampled profiles, as part of the preparatory stratigraphic mapping). Evidence from these projects has indicated that distances of 10–20 km are too large for reliable correlation of polarity zones between profiles: the rates of lava emplacement in the Iceland Tertiary appear to be quite variable with time and location, and the common petering out of individual lavas seen in cliff sections in NW Iceland indicates that the value of 1000 for the above ratio may be an overestimate in some cases. However, in view of the scarcity of feeder dykes this value is unlikely to be less than 100.

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The well-exposed and regular lava sequences of NW Iceland are a good place to test to what extent the palaeomagnetic polarity pattern of closely spaced profiles agrees or disagrees. Also, evidence of more and more short geomagnetic reversal events has been accumulating during the last few decades, not only in the Icelandic lava pile but also elsewhere: thus, the 'Jaramillo event' of around 1 Myr in age is now accompanied by at least four other subchrons (Singer & Brown 2002). Such observations call for increased attention to details in correlation. An additional motive for these palaeomagnetic studies in NW Iceland concerns the progressive change in long-term secular variation with time between approximately 15 and 1 Myr ago originally reported by Kristjansson & Johannesson (1989) and supported by subsequent studies (Kristjansson 1995, 2002). Continued work on lava sequences in Iceland may help understanding changes in the stability of the geomagnetic field since 15 Myr ago and thus, for example, improve resolution in the interpretation of oceanic anomaly lineations.

2.3 Palaeomagnetic sampling and measurements

Sampling for palaeomagnetic laboratory measurements in the profiles listed in Table 1 and shown in Figs 2–4 was based on stratigraphic mapping by BSH in 1997–2001. In some of the profiles studied in detail, assistance in this mapping was provided by eight participants in an undergraduate geological field course in September 2000. The sampling was carried out by LK in 1998 (profile KE) and 1999 (GD, TF, most of ND), LK and HA in 2000 (ST/SS, HN, GA, SV, GE, GF, TH, TI, NC/ND/NE) and 2001 (GO, FF, FE, TO, SW). Sediment sites were sampled by HA in 2001. In most cases, outcrops cannot be traced continuously between these profiles, as they are separated by fjords, inlets, valleys, scree-covered hillsides, etc. A profile KO at the southernmost outcrop of old lignite sediments (Fig. 1) has not been sampled. In general, four 25 mm core samples were collected per flow, spread over a few to several metres laterally. They were oriented in azimuth by sighting on the Sun or on distant objects of known position. The main exception is flows SV 1–12 where the only object available was a small reef not seen on maps or aerial photographs; in this case an estimate of the mean local magnetic declination has been used.

The remanence of one specimen from each core sample was measured at the University of Iceland, using an Institut Dr Förster four-probe static fluxgate magnetometer. Measurements were made of the natural remanence and following AF demagnetization at 10, 15 and 20 mT peak field in a Molspin two-axis tumbler demagnetizer. Viscous magnetization is generally removed by 10 mT treatment and almost always by 15 mT. AF treatment at 25 and occasionally 30 mT was carried out on a sample if its direction change between 15 and 20 mT exceeded 2°. Only two flows appear to have been affected by lightning, NE 18 and FF 8. Extended demagnetization was applied to several specimens, and results from five of these from north of Isafjardardjup are shown in Fig. 5. In general, instability may set in at peak fields exceeding 15–20 mT in material having a weak or soft remanence. Minor amounts of rotational remanence (RRM) have been noted in some of the present lava samples AF-demagnetized to 40 mT or beyond. The RRM is manifested as progressive systematic deviations (from a stable direction obtained below 40 mT) with increasing peak field. Its effects are easily eliminated by averaging two directional measurements, following demagnetizations to the same peak AF with the sample rotating in an opposite sense.

Low-field thermomagnetic curves were obtained (in air) from 13 samples, using a Bartington MS2W furnace meter. These curves

Table 1. List of stable remanence directions and virtual pole positions in lava flows from the current survey. All directions have been corrected for regional tectonic tilt of 2° or less to the southeast. Dec., Inc. = declination (east of north) and inclination (down) of mean direction. Long., Lat. = east longitude and north latitude of the corresponding virtual pole. Alf = 95 per cent confidence angle for the mean direction. J100 = arithmetic mean remanence intensity after 10 mT AF treatment. Pol. = polarity (N = normal, R = reverse, with T and E indicating latitudes numerically less than 40° and 10°, respectively, *indicating that alf > 18°).

Flow	N	Dec.	Inc.	Long.	Lat.	Alf	J100	Pol.
	SV KOGUR–FLJOTAVIK							
SV 1	3	14	+84	349	+77	2	0.19	N
SV 2	3	88	+59	50	+37	3	0.64	NT
SV 3	3	86	+62	50	+40	3	0.78	N
SV 4	3	339	+72	213	+77	5	0.80	N
SV 5	3	278	+67	267	+48	9	1.88	N
SV 6	3	127	+64	16	+29	11	0.33	NT
SV 7	3	194	−49	316	−53	5	0.57	R
SV 8	3	187	−73	311	−81	4	2.42	R
SV 9	3	184	−72	325	−81	4	4.14	R
SV10	3	169	−52	353	−56	2	2.09	R
SV11	4	251	−86	178	−68	2	0.82	R
SV12	4	276	−79	198	−57	3	3.26	R
SV12X	4	175	−78	102	−88	3	0.79	R
SV13	4	176	−70	348	−77	3	0.88	R
SV14	4	75	+35	73	+24	5	0.39	NT
SV15	4	76	+32	73	+21	3	0.44	NT
SV16	4	175	−61	347	−66	4	1.77	R
SV17	4	221	−76	240	−72	7	5.17	R
SV18	4	201	−79	228	−82	3	4.99	R
SV19	4	201	−78	238	−82	4	3.75	R
SV20	4	202	−76	258	−80	2	8.04	R
SV21	4	204	−77	245	−80	3	4.43	R
SV22	4	185	−28	331	−38	25	0.43	RT*
SV23	4	177	+34	340	−5	7	0.65	E
	GA GARDAFJALL							
GA 1	4	23	+70	103	+74	2	3.75	N
GA 2	4	100	+78	19	+55	7	1.23	N
GA 3	4	163	+73	348	+36	2	3.32	NT
GA 4	4	25	+76	73	+79	2	3.43	N
GA 5	4	46	+81	27	+73	4	13.71	N
GA 6	4	256	+83	308	+60	3	4.00	N
GA 7	4	23	+80	32	+81	4	6.52	N
GA 8	3	101	−41	67	−26	10	1.01	RT
GA 9A	4	42	+76	54	+73	3	3.62	N
GA 9B	4	20	+78	54	+82	3	4.56	N
GA 9C	Thin flow, omitted							
GA10	4	311	+81	289	+72	1	2.82	N
GA11	4	309	−86	170	−61	4	5.04	R
GA12	4	201	−84	178	−77	3	0.75	R
GA13	4	190	−69	312	−75	4	1.92	R
GA14	4	67	−84	134	−60	4	3.53	R
	ST/SS STRAUMNESFJALL							
ST 1	4	246	+77	301	+50	3	1.34	N
ST 2	4	208	+88	333	+63	5	3.96	N
ST 2A	4	209	+87	331	+62	2	4.73	N
ST 3	5	40	+81	26	+75	4	1.93	N
ST 4	4	38	+76	57	+74	2	4.33	N
ST 5	Inaccessible							
ST 6	3	268	−39	240	−21	5	0.69	RT
ST 7	3	284	+49	247	+33	3	0.83	NT
ST 8	4	268	+71	278	+48	4	1.95	N
ST 9	4	327	+70	224	+69	3	1.44	N
ST10	4	248	−85	183	−68	1	5.02	R
ST11	4	172	−69	357	−75	3	2.05	R

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Table 1. (*Continued.*)

Flow	N	Dec.	Inc.	Long.	Lat.	Alf	J100	Pol.
ST12	4	184	−82	166	−82	4	1.70	R
SS 1	4	151	−35	14	−39	4	1.26	RT
SS 2	4	154	−85	139	−75	1	2.22	R
SS 3	4	126	−80	103	−70	3	1.53	R
SS 4	4	126	−83	123	−71	3	3.71	R
SS 5	3	282	−85	179	−63	3	1.51	R
SS 6	4	270	−80	198	−60	2	5.56	R
ST13	3	213	−70	269	−70	9	0.83	R
ST14	4	249	−70	233	−56	4	1.40	R
ST15	4	253	−67	235	−51	4	3.43	R
ST16	4	101	−86	139	−67	6	4.28	R
ST17	4	143	−87	145	−71	2	5.32	R
ST18	4	221	−14	291	−24	10	0.87	RT
ST19	4	221	−9	292	−22	5	1.30	RT
ST20	4	12	−85	153	−56	1	0.97	R
ST21	3	299	+65	246	+53	12	1.94	N
ST22	4	306	+66	240	+57	4	3.34	N
ST23	4	302	+65	244	+54	2	6.56	N
ST24	5	331	+76	245	+77	4	3.38	N
	HN HVARFNUPUR							
HN 1	4	18	+76	78	+82	5	3.00	N
HN 2	4	350	+69	183	+76	5	2.98	N
HN 3	4	338	+84	317	+77	3	1.54	N
HN 4	4	341	+83	316	+78	2	2.92	N
HN 5	4	338	+72	215	+76	3	1.48	N
HN 6	4	Scattered N and R					0.28	?
HN 7	4	204	−66	292	−68	6	0.73	R
HN 8	4	264	−79	204	−60	4	6.22	R
HN 9	4	264	−68	225	−47	32	2.86	R*
HN10	4	179	−68	340	−75	4	2.58	R
HN11		Inaccessible						
HN12		Inaccessible						
HN13	4	184	−44	332	−49	4	1.35	R
HN14	4	162	−52	4	−54	8	0.36	R
HN15	4	158	−54	10	−55	4	0.24	R
HN16	4	154	−47	13	−48	6	4.58	R
HN17	4	159	−53	8	−55	6	0.37	R
HN18	4	190	−78	233	−86	6	3.10	R
HN19	4	250	−80	207	−65	2	1.42	R
HN20	4	253	−81	202	−65	3	2.54	R
HN21	4	182	−57	334	−62	4	0.82	R
HN22	4	75	+45	69	+30	4	0.39	NT
HN23	4	239	−60	256	−49	4	1.62	R
	KE KERLING–SKALAVIK							
KE 0	5	169	+56	346	+14	7	0.67	NT
KE 1	4	348	+81	305	+83	3	1.39	N
KE 2	5	25	+76	73	+79	3	3.07	N
KE 3	4	287	+73	269	+57	3	5.08	N
KE 4	2	358	+75	169	+86	(11)	1.50	N
KE 5	5	9	+78	53	+86	2	2.06	N
KE 6		Thin, maybe top of KE 5						N
KE 7	4	136	−79	96	−73	3	1.80	R
KE 8	4	18	+7	137	+26	6	1.45	NT
KE 9		Inaccessible						
KE10	5	320	+38	206	+38	5	0.58	NT
KE11	5	191	−56	319	−60	8	1.43	R
KE12	6	194	−21	319	−34	8	0.97	RT
KE13	4	192	−31	322	−40	7	1.62	RT
KE14	9	131	−69	61	−62	2	3.80	R
KE14B	4	108	−79	105	−64	3	2.99	R
KE15	4	204	−76	251	−79	4	1.85	R
KE16	4	173	−47	347	−52	2	4.38	R
KE17	5	226	−53	273	−48	2	2.47	R
KE18	4	202	−54	303	−56	5	3.32	R

Table 1. (*Continued.*)

Flow	N	Dec.	Inc.	Long.	Lat.	Alf	J100	Pol.
KE18A	4	174	−67	351	−74	4	1.42	R
KE19	5	157	−65	20	−68	5	4.94	R
KE20	4	247	−59	248	−45	3	1.56	R
KE21	5	79	+2	77	+6	3	0.95	E
KE22	4	184	−56	330	−60	3	1.20	R
KE23	5	219	−61	275	−58	8	0.46	R
KE24	4	223	−75	237	−71	1	4.41	R
KE25	4	156	−78	84	−80	3	5.89	R
KE26	5	35	−76	136	−42	4	1.28	R
KE27	6	1	−32	156	+6	5	1.49	E
KE28	5	334	+53	195	+54	2	5.42	N
KE29	6	357	+47	161	+52	6	0.99	N
KE30	4	Small outcrop, unstable					1.41	N?
KE31	4	345	+68	191	+73	2	1.29	N
KE32	4	0	+48	156	+53	1	4.80	N
KE33	4	115	+83	1	+58	4	4.29	N
KE34	8	109	−9	51	−12	6	0.88	RT
KE35	4	208	−69	278	−70	3	7.85	R
KE36	4	178	−28	339	−39	5	2.25	RT
KE37	5	199	−68	293	−73	4	4.18	R
KE38	5	136	+69	7	+33	4	1.63	NT
	GD STIGAHLID							
GD 1	4	140	−75	72	−72	2	9.97	R
GD 2	4	176	−66	346	−73	6	3.29	R
GD 3	5	206	−69	280	−71	3	2.58	R
GD 4	4	137	−73	65	−69	10	2.29	R
GD 5	4	336	−79	169	−46	4	4.20	R
GD 6	5	340	−77	168	−43	4	1.27	R
GD 7	4	242	−76	223	−65	3	4.54	R
GD 8	5	173	−51	347	−55	4	5.99	R
GD 9	4	162	−86	146	−75	4	3.56	R
GD10	5	160	−83	131	−79	5	8.07	R
GD11	4	334	−85	164	−57	6	3.25	R
GD12	5	138	−78	87	−73	3	5.21	R
GD13		Thin inaccessible flow						R?
GD14	4	231	−85	184	−71	2	2.53	R
GD15	4	202	−74	270	−78	6	2.32	R
GD16	4	181	−59	335	−63	6	4.97	R
GD17		Inaccessible						
GD18		Inaccessible						
GD19	4	251	+81	306	+57	4	2.20	N
GD20	4	153	−65	27	−66	7	5.49	R
GD21	4	159	−67	19	−70	3	5.13	R
GD22	4	250	−68	236	−53	12	3.01	R
GD23	4	269	−77	206	−56	5	2.21	R
	GE TRADARHORN							
GE 1	5	188	−67	319	−73	6	1.12	R
GE 2	4	207	−74	261	−77	5	1.74	R
GE 3	9	87	−38	78	−18	9	0.56	RT
GE 4	4	154	−78	81	−80	2	3.98	R
GE 5	5	187	−35	328	−43	5	1.61	R
	GF BOLUNGARVIK							
GF 1	5	182	−52	334	−57	3	4.68	R
GF 2	4	175	−54	344	−58	6	3.55	R
GF 3	3	Doubtful whether *in situ*						R?
GF 4	4	206	−71	277	−73	3	9.56	R
GF 5	4	211	−79	221	−78	5	2.85	R
GF 6	5	248	−72	231	−58	2	2.59	R
GF 7	4	261	−82	195	−64	2	10.15	R
GF 8	5	204	−66	290	−69	6	4.59	R
GF 9	4	173	−65	351	−71	2	1.66	R
GF10	5	179	−80	151	−85	2	7.11	R
GF11	5	167	−55	358	−58	9	3.38	R
GF12	4	352	−34	165	+5	10	1.36	E

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Table 1. (*Continued.*)

Flow	N	Dec.	Inc.	Long.	Lat.	Alf	J100	Pol.
GF13	4	10	+66	136	+72	3	6.08	N
GF14	4	26	+78	44	+80	3	7.30	N
GF15	4	342	+65	192	+69	4	8.01	N
GF16	4	15	+72	113	+78	3	3.10	N
GF17	4	307	+62	234	+53	4	1.57	N
GF18	4	54	+46	89	+39	4	0.79	NT
GF19	4	117	+53	30	+20	2	1.69	NT
GF20	4	356	+81	325	+83	4	8.47	N
GF21	4	349	+79	292	+85	3	13.61	N
GF22	4	15	+80	21	+83	6	15.13	N
GF23	1	One oriented sample						N
GF24	1	One oriented sample						N
GF25	1	One oriented sample						R
GF26	4	183	−70	328	−78	3	3.29	R
	NC/ND/NE NESDALUR–BARDI							
NC 1	4	213	−56	287	−55	4	0.35	R
NC 2	3	242	−62	248	−50	5	2.35	R
NC 3	4	198	−71	290	−76	4	3.04	R
NC 4	Inaccessible							
ND 1	4	177	−45	341	−51	3	2.20	R
ND 2	4	227	−70	251	−64	3	0.31	R
ND 3	4	305	−79	183	−51	5	0.78	R
ND 4	5	127	−78	93	−69	3	5.32	R
ND 5	4	176	−65	345	−70	4	5.05	R
ND 6	4	181	−70	332	−77	4	3.03	R
ND 7	4	258	−75	216	−58	2	6.84	R
ND 8	Omitted: Same flow as ND					9 ?		R?
ND 9	6	250	−74	224	−59	5	6.47	R
ND10	4	176	−73	352	−83	4	2.71	R
ND11	4	189	−74	298	−83	3	2.48	R
ND12	4	138	−79	94	−74	4	1.43	R
ND13	5	122	−82	113	−70	3	4.25	R
ND14A	4	209	−79	217	−79	3	2.88	R
ND15	4	226	−74	241	−68	5	4.57	R
NE 1	4	146	−56	27	−54	11	1.15	R
NE 2	4	203	−58	298	−60	7	1.98	R
NE 3	5	267	−65	225	−43	10	0.36	R
NE 4	4	303	−70	195	−37	22	0.63	RT*
NE 5	5	42	+55	96	+51	5	1.18	N
NE 6	4	21	+66	115	+70	5	1.05	N
NE 7	4	28	+71	92	+72	8	1.43	N
NE 8	4	8	+79	19	+86	4	3.24	N
NE 9	4	9	+74	117	+83	3	6.27	N
NE10	5	19	+54	127	+56	7	1.03	N
NE11	4	210	−62	286	−62	4	2.60	R
NE12	4	143	+42	10	+4	8	2.17	E
NE13	4	72	+60	62	+44	3	1.75	N
NE14	4	124	+86	349	+61	6	2.06	N
NE15	3	27	+50	118	+52	5	5.40	N
NE16	4	36	+52	105	+51	6	2.97	N
NE17	4	34	+54	107	+53	6	4.61	N
NE18	5	286	+85	311	+67	7	(6.08)	N
NE19	4	34	+74	70	+74	4	7.58	N
NE20	5	104	+80	11	+56	5	1.90	N
	TF HELGAFELL							
TF 1	6	191	−49	321	−53	6	2.19	R
TF 2	4	206	−68	283	−70	5	5.52	R
TF 3	4	211	−69	272	−69	9	2.53	R
TF 4	6	190	−65	315	−71	6	3.55	R
TF 5	2	167	−49	356	−53	(14)	3.54	R
TF 6	7	333	+53	196	+54	5	4.10	N
TF 7	4	74	+86	356	+67	3	4.23	N
TF 8	4	340	+67	199	+71	3	7.83	N

Table 1. (*Continued.*)

Flow	N	Dec.	Inc.	Long.	Lat.	Alf	J100	Pol.
TF 9	Omitted, poor outcrop							?
TF10	4	340	+63	192	+65	5	5.69	N
TF11	4	353	+61	169	+66	7	2.03	N
TF12	4	351	+66	176	+72	7	3.17	N
TF13	4	345	+64	186	+69	6	3.21	N
TF14	4	117	+86	350	+62	5	8.88	N
TF15	4	353	+85	331	+76	5	3.04	N
TF16	4	352	+60	171	+64	4	2.80	N
TF17	4	46	+45	97	+41	5	1.54	N
TF18	4	52	+53	86	+45	7	0.90	N
TF19	4	37	+48	106	+47	5	1.34	N
TF20	4	39	+44	105	+43	3	1.30	N
TF21	9	314	−65	190	−28	11	0.77	RT
TF22	5	10	+70	129	+77	2	8.51	N
TF23	5	8	+65	141	+70	5	10.04	N
TF24	4	354	+69	173	+76	4	10.59	N
TF25	4	323	+67	223	+65	3	4.01	N
TF26	4	335	+64	203	+66	4	3.71	N
	TH HOFN							
TH 1	4	345	+72	202	+79	4	0.97	N
TH 2	4	69	+88	345	+67	2	3.05	N
TH 3	4	159	+82	345	+51	5	4.85	N
TH 4C	4	329	+82	298	+77	2	8.48	N
TH 5	4	324	+64	218	+62	4	1.85	N
TH 6	4	Breccia, scattered dir.					1.02	N?
TH 7	4	10	+57	139	+61	7	2.41	N
TH 8	4	346	+70	193	+77	7	8.77	N
TH 9	4	323	+67	224	+66	4	4.95	N
TH10	4	337	+75	234	+79	8	4.73	N
TH11B	4	315	+71	242	+67	3	7.10	N
TH12	4	304	+67	245	+58	4	6.29	N
TH13	4	20	+64	118	+67	6	3.21	N
TH14	4	310	+52	224	+46	7	0.81	N
TH15	4	131	+66	12	+30	5	2.89	NT
TH16	4	291	+24	232	+20	12	1.12	NT
TH17	4	300	+29	225	+26	7	0.67	NT
TH18	5	234	+66	297	+32	6	1.86	NT
	TI HAFNARDALUR							
TI 1	6	216	+68	311	+31	6	0.64	NT
TI 2	6	163	+70	347	+31	7	0.60	NT
TI 3	4	6	+67	142	+73	3	7.83	N
TI 4	4	336	+77	250	+80	3	8.05	N
TI 5	4	335	+57	196	+58	6	4.00	N
TI 6	4	336	+54	195	+55	5	2.96	N
TI 7	4	40	+69	81	+67	4	7.03	N
TI 8	4	19	+77	62	+82	4	2.78	N
TI 9	4	326	+82	299	+76	6	6.84	N
TI10	4	300	+73	261	+63	3	3.14	N
TI11	4	326	+58	209	+57	3	3.20	N
TI12	4	18	+57	127	+60	5	1.82	N
TI13	4	1	+66	153	+73	8	2.95	N
TI14	4	358	+71	161	+80	4	3.37	N
TI15	4	346	+71	196	+78	5	3.50	N
TI16	4	357	+77	216	+88	3	4.44	N
TI17	4	10	+71	125	+79	3	6.86	N
TI18	4	328	+52	201	+51	8	5.60	N
	SW SKALAVIKURHEIDI							
SW 1	4	232	−75	232	−68	2	2.81	R
SW 2	5	175	−51	344	−55	3	3.20	R
SW 3	3	242	−49	259	−38	6	1.58	R
SW 4	4	181	−69	335	−76	5	1.28	R
SW 5	5	164	−67	11	−72	2	2.34	R
SW 6	Thin crumbly flow							R
SW 7	4	160	−67	18	−70	3	1.87	R

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Table 1. (*Continued.*)

Flow	N	Dec.	Inc.	Long.	Lat.	Alf	J100	Pol.
SW 8	5	231	−68	253	−60	10	2.02	R
SW 9	4	258	−58	238	−40	2	2.82	RT
SW10	4	181	−32	335	−41	15	0.43	R
SW11	4	77	−2	79	+4	10	0.93	E
SW12	4	183	−56	331	−60	1	5.89	R
SW13	4	179	−54	338	−58	2	1.35	R
	GO NORDUREYRI, N-SUGANDAFJORDUR							
GO 0	4	Unstable, scattered dir.					0.24	?
GO 1	4	310	+45	220	+40	4	1.25	N
GO 2	4	315	+56	221	+50	6	0.48	N
GO 3	4	62	−81	130	−55	6	0.40	R
GO 4	3	194	−77	249	−84	4	0.97	R
GO 5	4	153	−38	11	−42	2	4.76	R
GO 6	4	170	−50	352	−54	3	4.92	R
GO 7	4	188	−71	313	−79	3	8.97	R
GO 8	4	254	−62	239	−45	3	1.67	R
GO 9	5	234	−71	242	−63	3	2.87	R
GO10	5	233	−75	230	−67	3	3.45	R
GO11	4	85	−76	109	−54	6	2.88	R
GO12	5	140	−83	119	−74	3	6.45	R
GO13	4	167	−86	149	−75	4	10.77	R
GO14	4	183	−81	167	−85	2	3.64	R
GO15	4	187	−70	319	−77	5	3.52	R
GO16	4	173	−67	353	−74	4	2.40	R
GO17	4	173	−69	354	−76	3	1.90	R
GO18	5	169	−65	359	−70	4	2.54	R
GO19	5	175	−62	345	−67	3	2.25	R
GO20	4	195	−54	313	−57	5	2.08	R
GO21	1	One oriented sample						R
GO22	4	223	−71	252	−68	2	4.43	R
GO23	4	83	+2	73	+4	17	1.43	E
GO24	4	180	−68	337	−75	5	0.61	R
GO25	4	295	−22	219	−1	8	3.73	E
GO26	4	325	+61	213	+59	5	4.49	N
GO27	4	214	−56	286	−54	13	0.79	R
GO28	4	226	−57	271	−51	6	2.11	R
GO29	4	6	+64	144	+70	2	4.70	N
GO30	1	One oriented sample						N
	FE FLATEYRI, N-ONUNDARFJORDUR							
FE 1	4	148	−60	29	−59	5	6.01	R
FE 2	4	155	−62	19	−63	3	0.93	R
FE 3	4	104	−71	88	−54	3	3.49	R
FE 4	4	174	−76	17	−86	4	1.10	R
FE 5	4	227	−73	243	−68	5	5.46	R
FE 6	4	138	−88	148	−69	7	0.91	R
FE 7	5	182	−66	332	−72	3	3.30	R
FE 8	4	186	−24	329	−36	3	0.96	RT
FE 9	5	25	+72	92	+75	4	2.65	N
FE10	5	324	+80	282	+76	4	2.28	N
FE11	4	127	+87	345	+63	3	3.03	N
FE12	4	19	+43	130	+47	2	1.03	N
FE13	4	20	+57	124	+59	4	0.97	N
FE14	4	116	+64	25	+32	5	1.59	NT
FE14A	1	One oriented sample						N
FE15	5	1	+51	154	+56	6	1.70	N
FE16	4	300	+21	223	+22	8	0.53	NT
FE17	4	183	−80	167	−85	6	1.87	R
FE18	4	192	−36	322	−43	2	0.74	R
FE19	4	212	−75	247	−76	22	0.82	R*
FE20	4	217	−52	284	−50	4	3.02	R
FE21	4	186	−41	328	−47	7	0.79	R
FE22	4	175	−56	345	−61	8	3.07	R
FE23	5	159	−33	3	−40	7	0.37	R

Table 1. (*Continued.*)

Flow	N	Dec.	Inc.	Long.	Lat.	Alf	J100	Pol.
FE24	4	203	−44	306	−47	5	0.75	R
FE25	3	241	−29	267	−26	5	0.97	RT
FE25A	1	One oriented sample						R
FE26	4	144	−65	38	−63	6	1.57	R
FE27	4	19	+82	5	+80	2	1.70	N
FE28	4	11	+80	16	+84	6	2.75	N
FE29	4	123	+56	23	+21	3	0.36	NT
FE30	4	303	+58	235	+48	5	1.83	N
FE31	4	347	+32	173	+41	4	3.01	N
	FF N- ONUNDARFJORDUR							
FF 1	4	255	−69	230	−52	3	5.37	R
FF 2	3	246	−58	250	−44	11	2.67	R
FF 3	Inaccessible							
FF 3A	4	235	−69	247	−60	4	0.98	R
FF 4	4	206	−85	175	−75	2	2.26	R
FF 5	4	158	−70	30	−74	5	2.79	R
FF 6	5	38	−87	149	−61	1	4.85	R
FF 7	4	109	−77	98	−62	2	14.33	R
FF 8	4	157	−55	12	−57	31	(25.13)	R*
FF 9	4	194	−72	294	−79	4	4.76	R
FF10	4	151	−67	34	−68	4	4.12	R
FF11	4	186	−64	324	−70	3	2.34	R
FF12	3	181	−63	334	−68	6	3.28	R
FF13	4	189	−61	320	−65	3	2.60	R
FF14	4	258	−72	222	−54	6	1.56	R
FF15	4	273	−74	209	−52	5	0.75	R
FF16	3	208	−79	219	−79	4	1.88	R
FF17	4	148	+81	351	+50	5	5.38	N
FF18	4	344	−39	172	+1	9	1.87	E
FF19	4	194	−61	311	−65	8	2.26	R
FF20	4	193	−49	318	−53	2	2.53	R
FF21	4	232	−60	261	−52	5	0.66	R
FF22	4	276	−80	194	−58	5	4.26	R
FF23	4	150	−59	24	−58	3	1.56	R
FF23C	1	One oriented sample						R
FF24	Poor outcrop							
FF25	4	184	−62	329	−67	1	4.60	R
FF26	4	83	+69	44	+49	4	3.86	N
FF27	4	118	+79	8	+51	4	5.02	N
FF28	4	38	+48	106	+46	2	1.15	N
	TO MT. THORFINNUR, S-ONUNDARFJORDUR							
TO 1	4	183	−63	330	−69	2	1.79	R
TO 2	4	181	−58	335	−63	2	1.80	R
TO 3	4	175	−55	344	−59	2	2.46	R
TO 4	4	187	−65	322	−71	3	11.61	R
TO 4A	5	139	−77	83	−73	4	5.67	R
TO 4B–4D	Inaccessible							
TO 4E	4	100	−12	60	−10	13	0.59	E
TO 5	8	337	−60	175	−18	6	1.07	RT
TO 6	4	336	−59	176	−17	7	0.28	RT
TO 7	4	199	−75	263	−81	3	4.73	R
TO 8	4	195	−63	307	−67	8	2.06	R
TO 9	4	177	−55	341	−59	4	1.85	R
TO10	4	146	−70	46	−69	6	1.62	R
TO11	4	146	−68	43	−67	4	1.47	R
TO12	4	160	−45	4	−48	2	1.14	R
TO13	4	167	−47	355	−51	14	1.76	R
TO14	5	166	−55	359	−59	4	3.49	R
TO14A	4	173	−57	349	−62	3	3.18	R
TO15	5	198	−74	273	−80	5	6.86	R
TO16	4	126	−59	52	−50	23	1.41	R*
TO17	4	56	−60	113	−25	6	0.90	RT
TO18	3	265	−46	239	−27	4	0.75	RT

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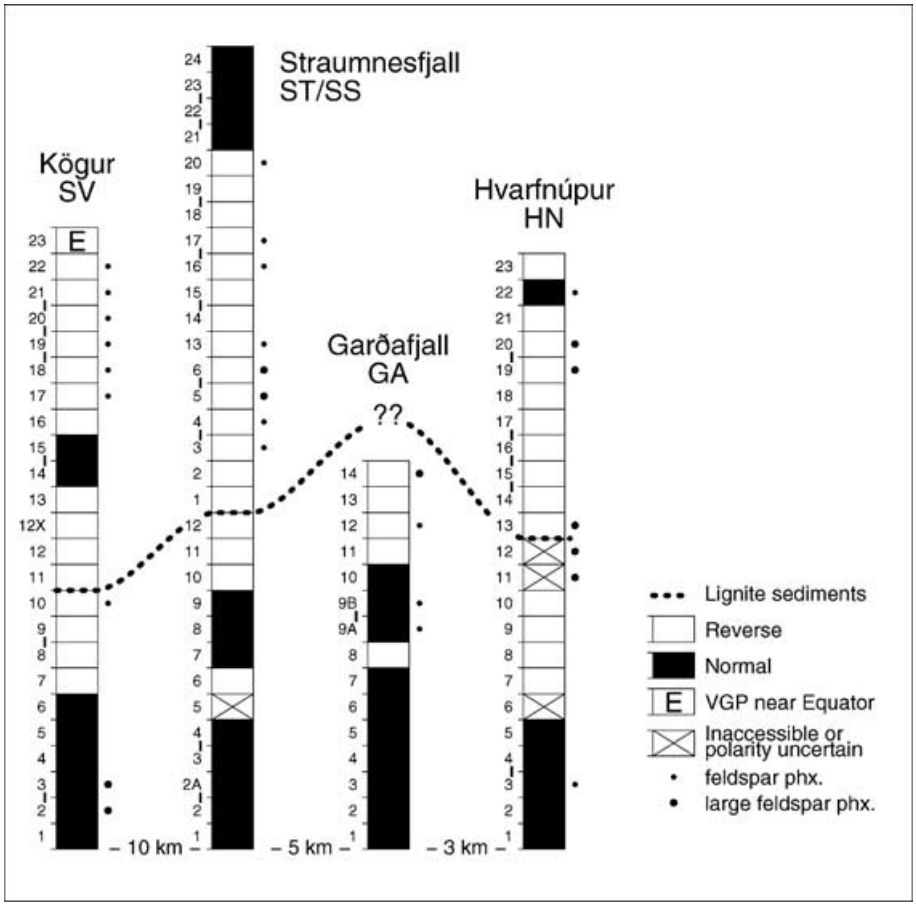


Figure 2. Polarity zones north of Isafjardardjup. All four profiles begin at sea level. They do not reach the tops of the respective hills which are approximately 400 m in height. The lignites and associated sediments occur at approximately 100–150 m altitude. A few olivine tholeiite lavas occur, e.g. GA 6, SV 4–6 and HN 4; additionally, some flows in the upper parts of the profiles (and above them) are ankaramitic texture, especially in profile SV. SV is in two segments approximately 0.4 km apart, SV 1–12 and 12X-23. Flow SV 12X is approximately at the same level as 12 but they have significantly different remanence directions so both are included here. The flows ST 1–24 were mapped in 1999 and the separately numbered flows SS 1–6 which are difficult to reach, were added in 2000. Short vertical lines between successive lava numbers indicate those that have similar remanence directions.

Table 1. (*Continued.*)

Flow	N	Dec.	Inc.	Long.	Lat.	Alf	J100	Pol.
TO19	5	216	−62	279	−60	3	0.79	R
TO20	5	212	−56	288	−55	4	6.37	R
TO20A-20D		Thin flow units, omitted						
TO21	4	195	−62	310	−65	1	3.62	R
TO22	4	156	−62	18	−63	1	2.93	R
TO23	4	166	−64	3	−69	3	2.70	R
TO24	5	292	+54	243	+40	29	4.32	N*
TO25	4	38	+43	107	+43	4	1.11	N
TO26	4	75	+70	48	+53	5	0.96	N
TO27	4	20	+54	126	+56	5	2.45	N
TO28	3	286	+16	235	+14	6	1.19	NT
TO29	4	Scattered directions					2.04	N?
TO30	4	347	+66	184	+71	7	3.46	N
TO31	5	16	+78	46	+83	6	2.14	N
TO32	3	138	+69	5	+33	3	1.13	NT
TO33		Steep crumbly outcrop						
TO34	5	187	−83	166	−80	2	4.10	R

were found to have a wide variety of shapes, of which two fairly typical examples are given in Fig. 6. In some samples there is a significant component having a Curie point in the range 180–280 °C, while others only exhibit Curie points of 500 °C or higher. The main conclusion that may be drawn from thermomagnetic curves in these and other Tertiary lavas from Iceland, is that the curves tend to have complex shapes in samples having a soft and unstable remanence. Essentially all the above curves are irreversible: in those where the room-temperature susceptibility does not change much, an increase in susceptibility (presumably by exsolution) upon moderate heating is balanced by a partial destruction of the main magnetic component (magnetite or maghaemite) at high temperatures.

Remanence intensities in the present collection (after 10 mT AF treatment, excluding lightning strikes and those flows where only one core was obtained) have an arithmetic average of 3.1 A m^{-1}, and a geometric average of 2.3 A m^{-1}. This is similar to values from elsewhere in the Tertiary lava pile of Iceland (Kristjansson 2002, fig. 3); the frequency distribution of intensities in these lavas is hyperbolic (Kristjansson & McDougall 1982). Room-temperature susceptibility was measured in one sample from each lava flow. The susceptibility values have an arithmetic average of slightly less than 0.02 SI units, comparable to averages of around 0.02–0.03 SI, which are found in some other Tertiary lava sequences in Iceland (Kristjansson 1984, 1999). They do not seem to correlate in any marked fashion with, for example, the intensity of natural remanence (*cf.* Kristjansson 1999, fig. 3) or the directional stability of the characteristic (primary) remanence component during AF treatment.

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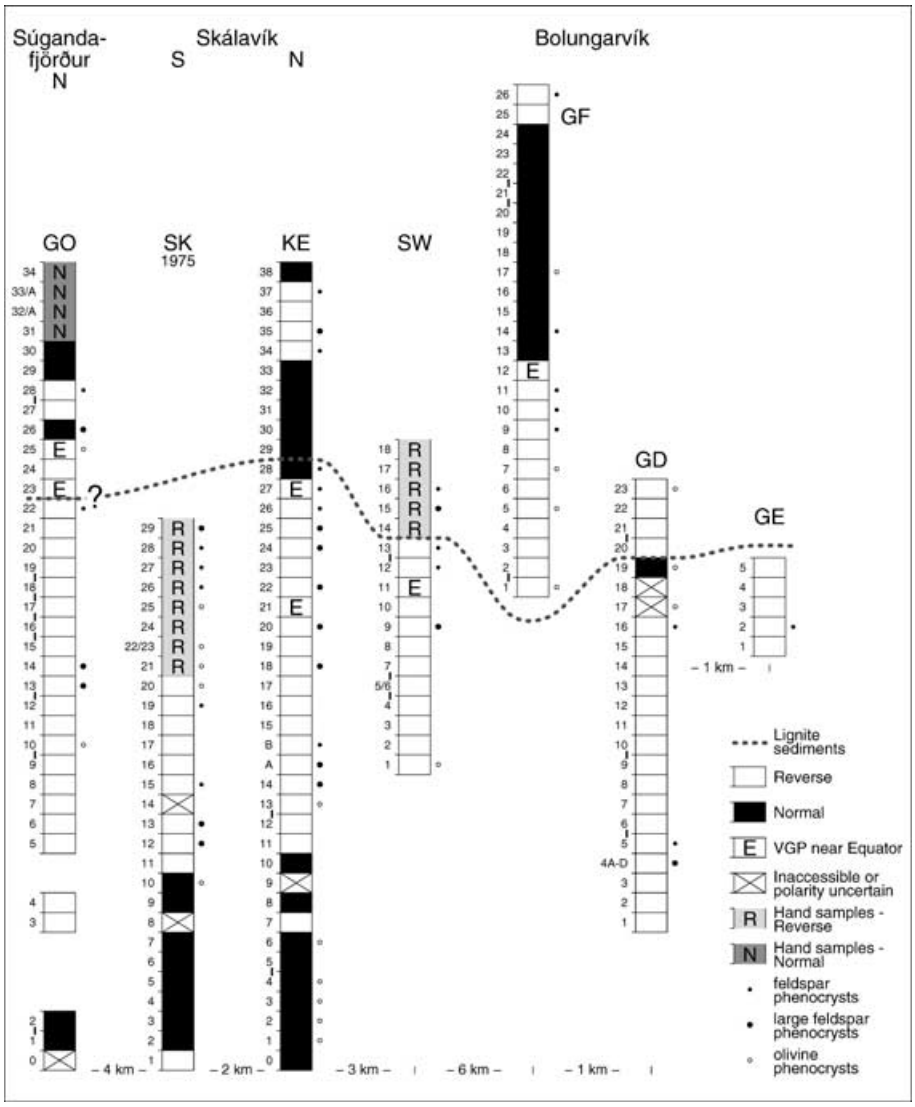


Figure 3. Polarity zones between Isafjardardjup and Sugandafjördur. Legend as in Fig. 2. The hills sampled in Figs 3 and 4 reach 500–700 m. The height of the lignites is in the range 150–300 m north of Dyrafjördur. Those profiles which begin above sea level, are elevated in the figure by approximately one lava per 10 m. The short profile GE was added to compensate for the inaccessible flows GD 17–18. The results in Table 1 for GD 12, SW 5 and KE 5 each include one sample from overlying thin flows.

3 RESULTS OF REMANENCE MEASUREMENTS

3.1 General

After each AF treatment, a mean direction and its 95 per cent confidence angle α_{95} (Fisher 1953) were computed for each lava flow. The mean having the lowest α_{95} was selected as the best one for that flow. In cases where two treatments gave very similar confidence angles (within 10 per cent, *cf.* McDougall *et al.* 1984) the mean direction from the higher peak field was selected. The means were corrected for the general tectonic tilt before computation of virtual geomagnetic pole positions (VGPs). Data for all lavas are listed in Table 1. The only flows where no reliable polarity was obtained were HN 6, GO 0, TO 29 and TH 6. Data from KE 30 and GF 3 are also not reported as the outcrops were small and possibly not in place. Eight single samples were discarded in the four profiles north of the Isafjardardjup fjord due to instability, possible orientation errors or other problems. Some resampling was undertaken where results were unsatisfactory, especially in 12 flows in profile KE; a few extra samples were also collected in profiles GE, TF and TI. 10 single samples were discarded in profile KE and altogether 22 in our other profiles south of Isafjardardjup. In those 404 lavas which have $\alpha_{95} < 18^\circ$, the average α_{95} is less than 5°.

Tight grouping of remanence directions in successive lava flows is fairly common in the profiles presented here. Various methods of detecting such groupings may be envisaged, all having a more or less arbitrary basis and being affected by many uncertainties including errors of measurement. We have chosen to use the following simple test criterion: any two flows are suspected of being erupted in rapid succession if the angular distance between their VGPs is less than 18°–(L/10), where L is the numerical value of their average VGP latitude, in degrees. The increase in the critical distance at low latitudes reflects among other things our view that the virtual pole is likely to move relatively rapidly there due to a high content of non-dipole components in the field (see, e.g., the random-walk model

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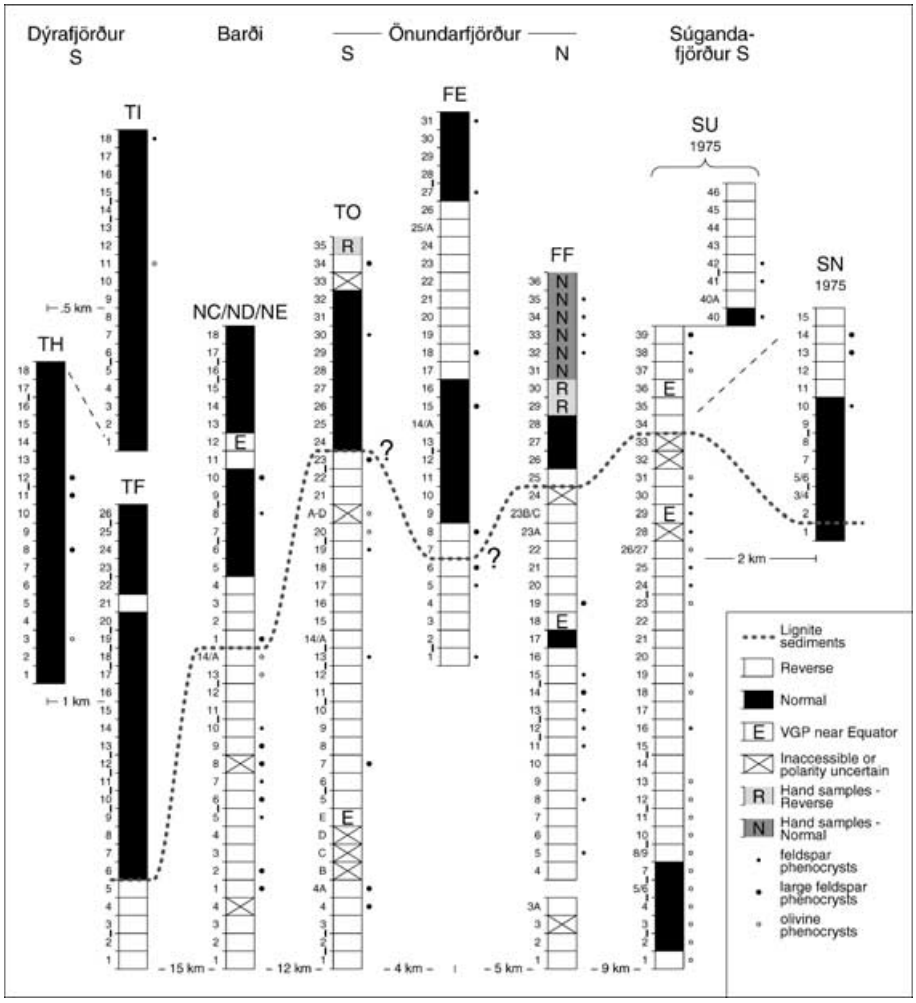


Figure 4. Polarity zones south of Sugandafjördur. Legend as in Fig. 2. Only a part of profile SU (sampled in 1975) is shown; it consists of two segments, with approximately 7 km distance between SU 39 and 40. Some thin flows such as SN 3 and 4, TO 20A-20D have been combined. Polarity data for SU and SN are from McDougall *et al.* (1984). The unexposed gap between flows FF 3A and 4 may contain sediments and weathered lavas described by Tryggvason (1960) at localities lying a few km to the northwest. Tryggvason correlated these with the lignite sediments but they seem to be at a lower stratigraphic level. Flows TO 4B–4D were too steep to sample but field measurements indicate that they are of reverse polarity. The profile NC/ND/NE is essentially in one piece but a couple of flows may be hidden above NC 4. Lignite-bearing sediments are not seen in profiles TO, GO and FE in spite of generally good exposures; their position is estimated. The top of TF is approximately at the same stratigraphic level as the bottom of TH but it is not possible to correlate them definitely as the flows are thin here.

of Kristjansson 1985). As 6° century^{-1} is a representative value of the movement of the VGP when in high latitudes (Mankinen *et al.* 1985), we consider that the directionally grouped lavas have in most cases been emplaced within a few centuries of each other. The lavas having grouped directions are identified with short vertical bars in Figs 2–4. The long-term mean interval between flows is expected to be of the order of 5–10 kyr as stated in Section 1.1 above.

3.2 Reversals and excursions north of Isafjardardjup (Fig. 2)

It is evident from Fig. 2 that the pattern of magnetic polarities is fairly similar in the four sampled profiles in Adalvik (ST/SS, GA, HN) and Kögur (SV). The bottom five to ten lava flows are normally magnetized and then a mostly reverse zone extends up to well above the lignite sediments. Grouping of successive paleomagnetic directions usually only involves two flows at a time (Fig. 2), the main exceptions being SV 18–21, ST 21–23 and HN 14–17.

Tentative correlations between the profiles can be made on the basis of details in the palaeomagnetic records, such as:

(i) an excursion or short reversal occurs in flows SV 2–3, ST 6–8 and GA 8;

(ii) another excursion in SV 14–15 may correspond to the low-latitude reverse poles seen, for example, in SS 1 and HN 13–17;

(iii) the low-latitude poles in SV 22–23 may date from the R–N transition between ST 20/21 and HN 21/22.

As mentioned above, Walker (1959) pioneered the stratigraphic use of lava groups having easily recognizable features such as abundant feldspar and olivine phenocrysts. However, it should be noted that evaluation of the petrographic character of a lava in the field or in hand samples and cores is a subjective one. Besides, units of intermediate character are common and the appearance of a lava often changes somewhat, both laterally and vertically. We had hoped that distinctive lava flows might be useful in the stratigraphical correlation in this area, especially flows with large feldspar crystals

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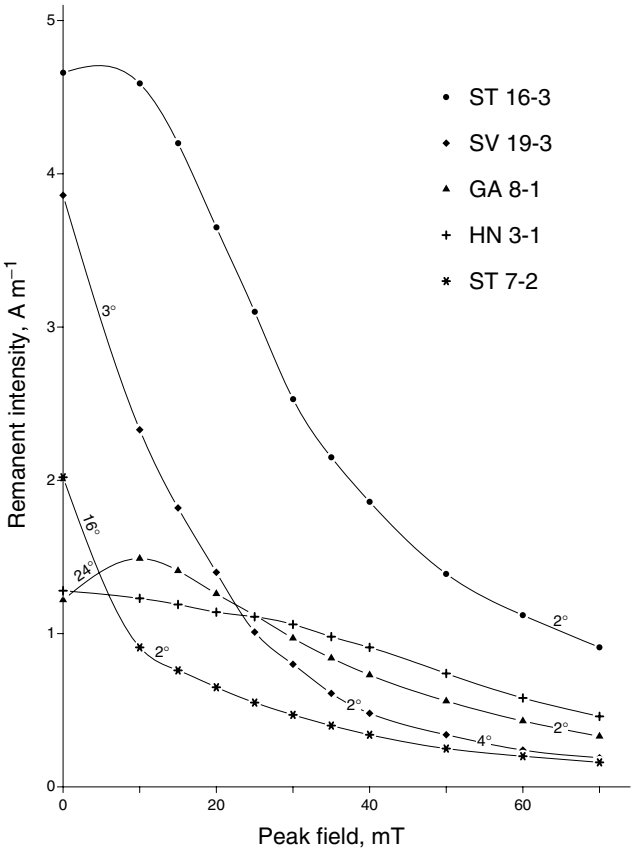


Figure 5. Results of extended alternating field demagnetization on five samples from lavas north of Isafjardardjup. The directional change between demagnetization steps is shown where it is 2° or greater. The values 3°, 16° and 24° occur where Brunhes age viscous magnetization is being removed; direction changes between other steps are random.

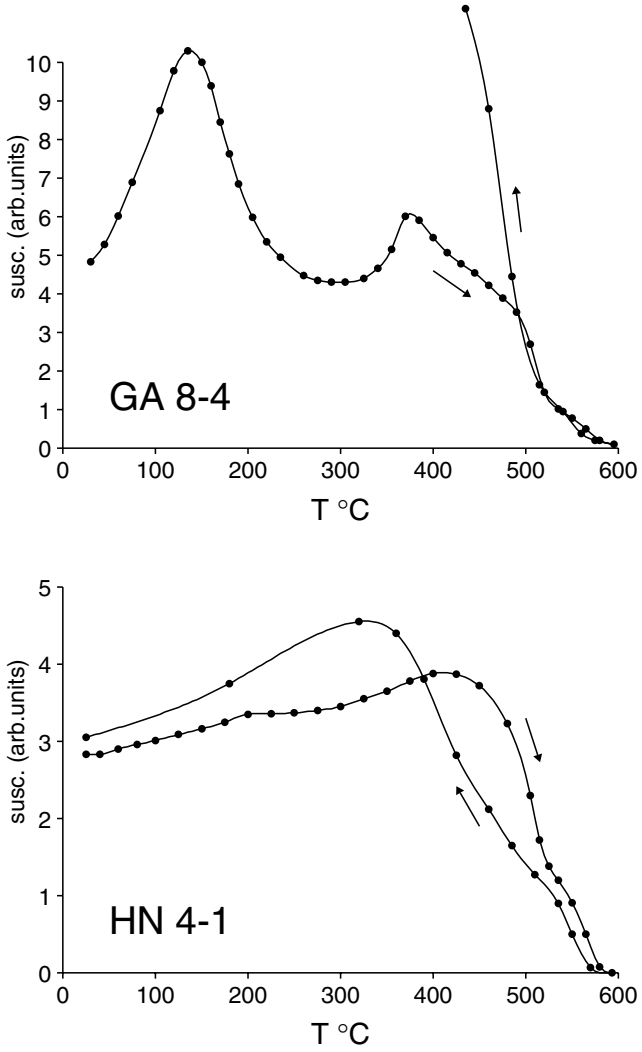


Figure 6. Two representative low-field susceptibility versus temperature curves (in air) from lava samples north of Isafjardardjup. The curve for GA 8–4, where a component with an apparent Curie point around 250 °C is gradually being transformed into fairly pure magnetite on heating, is not uncommon in samples carrying a soft or unstable remanence. The remanence of sample HN 4-1 is very stable to AF treatment.

(*cf.* Kristjansson *et al.* 1975). The presence of feldspar-porphyritic flows is indicated by dots in Fig. 2; however, little regularity can be seen in the distribution of these flows.

3.3 Reversals and excursions south of Isafjardardjup: Bolungarvik–Skalavik–north side of Sugandafjördur (Fig. 3)

Some grouping of directions in adjacent flows occurs in these profiles. As in the previous section, usually only two flows are involved at a time, except in GF 20–22, SW 4–7 and GO 15–19. With the very regular layering seen in the cliffs of the Skalavik–Bolungarvik area, we had anticipated the lava sequences in profiles SK and KE to be quite similar to those in GD, GE, GF approximately 10 km away. However, inspection of Fig. 3 reveals the occurrence of some eight lavas with large feldspar phenocrysts in profile KE, whereas a total of only one such lava is found in profiles GD, GE and GF. Also, 10 out of the 38 flows measured in KE have VGPs in latitudes of ±40° or less, but this is the case for only four out of 47 flows in GD, GE and GF. The normal-polarity zone at the base of the profiles KE, SK and GO is not seen in GD: it may possibly be hidden below the bottom accessible flows of that profile due to the general southeasterly dip. The normal-polarity zone KE 28–33 (containing the lignite sediments) seems to be absent in profile SW and is only represented by the flow GD 19 in profiles GD, GE and GF, plus possibly a few inaccessible flows below GD 19 and an excursion direction in GE 5. The excursion giving an equatorial pole in KE 21 may correlate with the low-latitude poles in SW 9–11, GE 3 and GO 23.

The distance from the Adalvik profiles across the Isafjardardjup fjord is approximately 20 km, so that full agreement between the polarity pattern in Adalvik and that in the Skalavik–Bolungarvik region is hardly to be expected even if the lignite-bearing sediments represent a single horizon. At any rate, the zone of reverse polarities below the lignites south of Isafjardardjup (Fig. 3) is much thicker than that in Adalvik and Kögur (Fig. 2).

3.4 Reversals and excursions: south side of Sugandafjördur–Önundarfjördur–Dyrafjördur (Fig. 4)

Grouping of more than two adjacent directions is seen in flows SU 4–7 and SN 4–6 (McDougall *et al.* 1984), FF 11–13, NE 15–17, TO 1–3, 12–14A, TF 9–13 (and possibly 17–20) and TI 13–15. There is no definite pattern to the distribution of feldspar-porphyritic and olivine-rich flows in the profiles. The flows below the lignite

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sediments are mostly of reverse polarity and those sampled above the sediments are largely normal.

Among possible stratigraphic correlations between profiles with the aid of details in the palaeomagnetic directions, we mention the following.

(i) The normal-polarity flows SU 2–7 may belong to the same subchron as GO 1–2, SK 2–10 and KE 0–6 (or 0–10). It should be noted that in their fig. 2, which is mostly based on field polarity measurements, Hardarson *et al.* (1997) correlated SU 2–7 with presumed normal-polarity zones in KE 22–24 and TF 1, which in fact are not present.

(ii) The single-flow excursion or polarity event in TF 21 (overlain by a few metres of a coarse, partly acidic, sediment) may correlate with that in NE 11–12.

(iii) The transitional direction in flow TH 18 is quite similar to that in TI 1 approximately half a km away downdip.

Above the sediments in profile TF there is evidence of rapid build-up of the lava pile. The upper part of TF (approximately 20 relatively thin flows) as well as profiles TH (18 flows) and TI (18 flows) are of normal polarity, with excursions occurring in TF 21 and in TH 14–18, TI 1–2. Several instances of grouped directions are noted in these profiles. The rapid build-up may be due to the fact that we are approaching a central-volcano complex (see McDougall *et al.* 1984, fig. 1), where the magma production rate is likely to be higher than in the areas between volcanoes.

3.5 Overall results on remanence directions and VGPs

In computing the mean remanence direction for the data of Table 1, one must first consider whether any flows should not be included. We have decided to leave out the flows KE 4 and TF 5 where only two samples were collected. We have also left out those seven lavas that have within-flow α_{95} values exceeding 18° due to instability or other problems: SV 22, HN 9, NE 4, FE 19, FF 8 and TO 16, 24. The cut-off value of 18° is more stringent than those used in many previous palaeomagnetic studies in Iceland, e.g. 23.5° by McDougall *et al.* (1984) and 20.5° by Kristjansson & Johannesson (1989). The choice of this rejection criterion is always somewhat arbitrary. However, errors of measurement are very small here compared with the 'true' between-flow scatter, so that changes in this criterion within wide bounds will have a trivial effect on the average fields. The mean direction which is given in Table 2, is slightly 'right-handed' with an inclination of 72°. This mean direction is shallower than that of the geocentric axial dipole field at this latitude, 77.5°. The mean field thus corresponds to a virtual pole which is 'far-sided' by several degrees. The far-sidedness is not primarily due to those lavas in the collection which give low-latitude VGPs: in Table 2 it is seen that if the 72 lavas having VGPs at latitudes ±45° or less are eliminated, the direction of the mean field changes by less than 2° of arc.

One may also ask whether some bias is introduced by the close grouping of remanence directions in successive lavas, pointed out above. In Table 2 we therefore give a mean field obtained after omitting 78 directions suspected of not being independent of those in lavas immediately preceding them. The resulting changes in the mean direction and statistical parameters are quite insignificant, as may be expected if the groupings occur at random in the lava pile. This is an example of the general robustness of the directional information contained in large palaeomagnetic collections from Icelandic lava flows.

Table 2 also includes mean VGP positions from the collection of Table 1. There are minor differences between the mean poles for normally and reversely magnetized lavas: the 232 reverse poles are less scattered but have a more far-sided mean than the 163 normal-polarity poles. It is possible that the period covered by the present collection is too short to average out some asymmetries of the field, especially if the reverse poles in our various profiles mostly represent a single-polarity chron.

Table 2. Average remanence directions and VGPs in the collection of Table 1 after rejection of those with $\alpha_{95} > 18°$ and inversion of all directions whose VGP latitude is <0°.

Fields	*N*	*R*	Dec.	Inc.	*k*	a.s.d.	α_{95}
All lavas	395	355.3	8.2	72.0	9.9	25.9	2.4
Lavas with VGP lat. >±45°	323	310.4	3.3	72.7	25.6	16.1	1.6
Only one lava per group	317	282.5	9.5	72.4	9.2	27.0	2.8
VGPs	*N*	*R*	Long.	Lat.	*k*	a.s.d.	α_{95}
All lavas	395	327.0	120	83.0	5.8	34.1	3.2
Normal lavas	163	129.0	118	86.5	4.8	37.8	5.7
Reverse lavas	232	198.4	121	80.7	6.9	31.3	3.8
Lavas with VGP lat. >±45°	323	292.5	139	84.1	10.6	25.1	2.5

N = number of units averaged. R = length of N unit vectors. Dec, Inc = declination and inclination of the mean field. k = Fisher's (1953) precision parameter. a.s.d. = angular standard deviation, calculated as arccos $[(R-1)/(N-1)]$. α_{95} = 95 per cent confidence angle for the mean. Long., Lat = longitude and latitude of the mean VGP.

Our mean pole may be compared with that of the composite section 10 in NW Iceland by Kristjansson & Johannesson (1989), which is at 83.9°N, 43°E, and to that of Kristjansson & Johannesson (1996) at 87.5°N, 165°E. The composite section 9 of Kristjansson & Johannesson (1989) contains the latter of these sections plus two profiles from McDougall *et al.* (1984), which do not influence the result greatly.

We see from this that palaeomagnetic poles from the western and northwestern parts of the northwest peninsula do have a tendency to be far-sided and/or right-handed, as noted already by Kristjansson (1968). The reasons for this apparent tendency are not known at present. Far-sidedness is not seen in results obtained by Kristjansson *et al.* (1995) from rocks of 10–13 Ma in East Iceland.

3.6 Trend in VGP scatter with age in Icelandic lavas

Inspection of the results of Table 1 reveals that there are a large number of virtual poles lying in middle and low latitudes. These poles occur at some of the polarity zone boundaries but also within the zones, indicating frequent instability of the geomagnetic dipole. Kristjansson (2002) recently published results that support previous conclusions by Kristjansson & Johannesson (1989) and Kristjansson (1995) regarding an apparent steady reduction with time in the angular standard deviation (a.s.d.) of virtual geomagnetic poles in Icelandic lavas. This makes it of particular interest to compute the a.s.d. of the poles in the present collection for comparison with fig. 3 of Kristjansson (2002). However, the data collection used in that study already included the results from our profiles KE, GD and TF (sampled in 1998–99). We therefore compute the angular standard deviation of VGPs in the other profiles of our Table 1, employing the same rejection criterion as Kristjansson (2002). The resulting a.s.d. from 314 flows is 34.2°, and it is shown in Fig. 7. The fact that it is lower than the a.s.d. from previous studies in the northwest peninsula, may indicate that the time interval covered by the present collection is not sufficiently long to average out asymmetries of the field on timescales of the order of 100 kyr, as suggested above.

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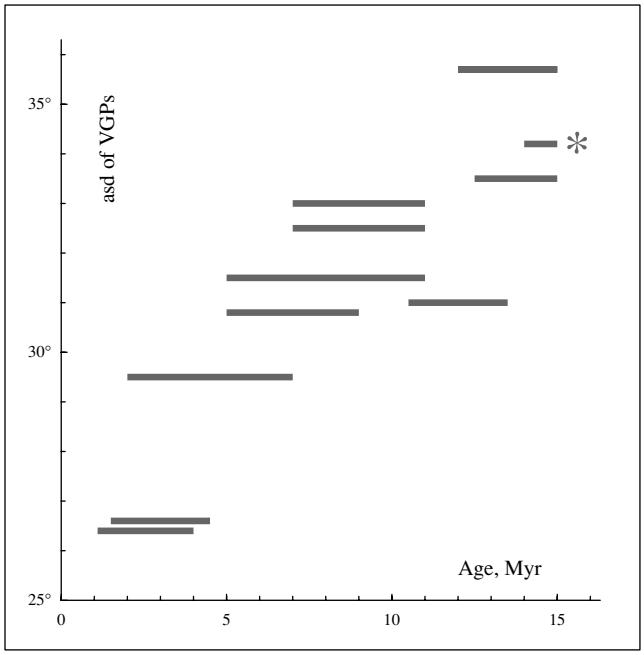


Figure 7. Between-flow a.s.d. of VGPs in various collections of Icelandic lava flows, in different regions and covering different estimated age ranges. Each data set represents from 300 to over 500 flows. The new data set marked with a star contains most of the present collection. The other data points are from Kristjansson (2002) with one minor change: data from profiles SF and SZ of McDougall *et al.* (1984) have been removed from the data set of the topmost point, because of relatively poor overall within-flow agreement in these. The 95 per cent confidence limits on each a.s.d. value, assuming Fisher's (1953) distribution and no serial correlation, are approximately $\pm 2^\circ$.

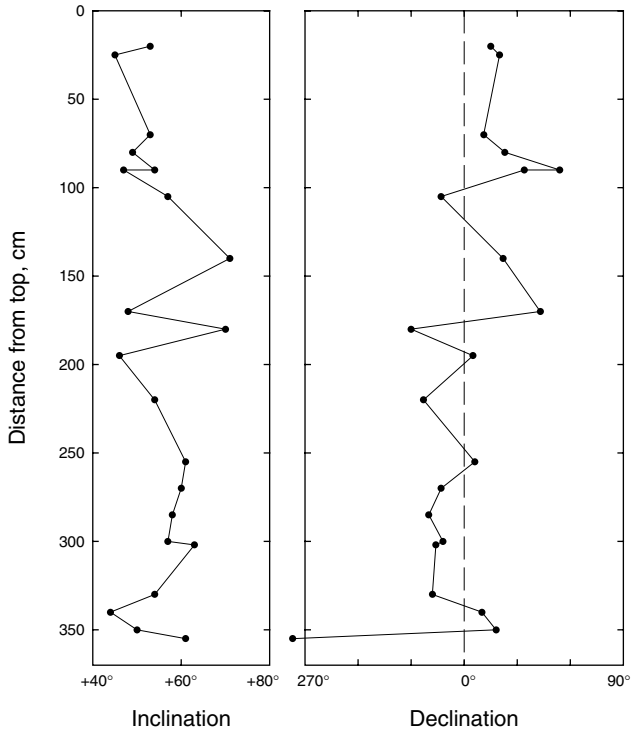


Figure 8. Remanence inclination and declination values in single samples from the 3.5 m thick lignite-bearing sediment layer between lava flows TF 5 and 6, Dyrafjördur. The directions which were obtained after 20 mT AF demagnetization, are stable and have an estimated error of 3°–4°.

3.7 Remanence intensities

Remanence intensity values (after 10 mT treatment) seem generally to be somewhat higher above the lignite sediments than below them. An increase in intensity with altitude has been found previously in some Tertiary lava sequences in Iceland at similar levels of alteration, while in others there is no appreciable change (Kristjansson 1984, fig. 4). However, this difference is difficult to quantify or to draw conclusions from regarding alteration effects in the present case, because intensities vary considerably between profiles. Thus, they are 1.5–3 times as high on average in the profiles GD, GE and GF in Bolungarvik than in SK, SU (McDougall *et al.* 1984) and KE to the west. Intensity values below the lignites in the northernmost profile SV of Fig. 1 are quite low, whereas in the nearby profile HN the mean intensity below the lignite sediments is higher than above them. This may reflect either variations in the chemistry of lava sources or differences in hydrothermal alteration.

3.8 Remanence of the lignite-bearing sediments

In an effort to estimate the length of time that it took the lignite-bearing sediments to form, we have collected numerous oriented samples at three sites in the major sedimentary layers of Figs 2–4. The layers are mostly composed of friable fine-grained sediment and tuff with minor lignite. Two of our sites are in profiles TF and SN, and one is approximately 5 km southeast of profile GE. The last-named site was also investigated by Roaldset (1983).

The lignite-bearing layer in profile TF in Dyrafjördur (Fig. 4) is 3.5 m thick. We collected 24 cores evenly distributed through the layer. Excluding three very weakly magnetized samples, the average NRM intensity is 0.3 A m^{-1}, which is comparable to values reported by Eiriksson *et al.* (1990) in thick sediments in North Iceland. The low-field susceptibility has an average of 0.011 SI. It shows a clear positive correlation with the natural remanence intensity (NRM) and with the amount of VRM acquired by the sediment samples over 2 weeks of storage. This VRM is of the order of 50 per cent of the NRM. Curie points in two samples are between 500 and 600 °C. During AF demagnetization, rotational remanence (RRM) appeared at around 20 mT; at 30 mT the intensity of the RRM was typically half of the surviving stable remanence. All the sediment samples had normal polarity (Fig. 8). An average direction of $D = 8^\circ$, $I = +57^\circ$, a.s.d. $= 15^\circ$, $\alpha_{95} = 6^\circ$ is obtained after exclusion of one anomalous sample from the base of the interbed and after correction for RRM but not for the small tectonic tilt. None of the samples showed any indication of reversed magnetization during the AF treatment. The overlying 8–10 m thick flow TF 6 has $D = 329^\circ$, $I = +55^\circ$ before tilt correction, $\alpha_{95} = 5^\circ$.

At profile SN in Sugandafjördur (Fig. 4) we collected 18 cores from the upper 2.2 m of the lignite-bearing sediment. It is approximately 3 m thick here (McDougall *et al.* 1984) but its lowest part is not accessible. Owing to high lignite content, only half of the samples had sufficient remanence intensity (>0.01 A m^{-1}) for reasonably accurate remanence measurements to be made after 20 mT AF demagnetization. All have normal polarity, the six most reliable ones giving a mean direction of $D = 4^\circ$, $I = +66^\circ$, a.s.d. $= 12^\circ$. Again this polarity agrees with that of the overlying flow which, however, has a rather unstable remanence (approximately $D = 5^\circ$, $I = +56^\circ$), according to unpublished measurements by L. Kristjansson in 1967 and by H. Audunsson from the present project.

At our third sediment site near profile GE, the interbed is divided by a 1.3 m thick lava, with 0.9 m of sediment below and 1.2 m above

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it. Cores from eight oriented hand samples were demagnetized to at least 20 mT. The NRM intensity is very variable, from 0.05 to 1.3 A m^{-1}. The samples all have reversed magnetic polarity, with scattered directions (averaging $D = 153^\circ$, $I = -63^\circ$, a.s.d. $= 37^\circ$) that are broadly similar to those in the underlying lava flow, the lava flow within the sediment and the thick overlying flow.

At the three sediment sites we cannot exclude the possibility that the sediment has been completely remagnetized by overlying flows. However, we do not find evidence for partial remagnetization (two remanence components of different directions) nor for a reversal occurring within the sediment at any of the sites. Currently it appears to us that the sediment now in place at each site was formed in a period much shorter than a geomagnetic chron.

4 CONCLUSIONS AND DISCUSSION

In 1998–2001 we collected approximately four oriented samples from each of 415 lava flows, for detailed palaeomagnetic measurements around a number of the oldest outcrops of lignite-bearing sediments in Iceland. The lava flows are generally excellent material for research on the geomagnetic field, as they have suffered little tectonic or geothermal disturbance. A stable primary remanence direction is easily isolated by AF treatment, and within-flow agreement is good. On the other hand, the fact that their thermomagnetic curves in air are generally irreversible indicates that they are not very suitable for bulk-rock palaeointensity determinations.

Our remanence measurements on these lavas (augmented by single samples from approximately 10 additional flows, data from 80 flows sampled in 1975, and some field observations) have been used to make the polarity columns shown in Figs 2–4. The frequent occurrence of two to three successive lavas having very similar directions, indicates that the lavas were emplaced at variable intervals, some shorter than a few centuries. These groupings appear to occur randomly in the sequences. The average time interval between flows was probably greater than 10 kyr, as some geomagnetic subchrons and excursions are not recorded in all profiles. The mean virtual pole in the present collection is 'far-sided' by 7°; possibly, the time interval covered by this collection was not sufficiently long for all variations (or persistent bias features) in the geomagnetic field to have been averaged out.

The lava collection includes various petrographic types, some flows being highly feldspar-porphyritic and others containing olivine phenocrysts. The fact that distinctive flows of these types cannot be traced reliably between profiles lying more than a few kilometres apart, indicates that they originate from a number of not very distant sources.

North of the fjord Isafjardardjup, our results (Fig. 2) are broadly compatible with the simple model that the rocks underlying the lignite sediments are all contemporaneous, and that those above the sediments are also contemporaneous.

South of Isafjardardjup (Figs 3 and 4) the lignites are generally underlain by a zone of reverse polarity but considerable lateral variations in the polarity zones occur. Various interpretations of the differences in the polarity pattern across Isafjardardjup may be advanced. One is that the polarity zones south and north of the fjord are the same but the lignite sediments on the south side are younger and therefore lie higher in the lava pile than north of the fjord, by 10 lava flows or so. Another interpretation is that the sediments are of the same age on both sides but the rate of emplacement of lavas (as well as the rate of subsidence) was considerably higher in the south. Still other interpretations are possible if one assumes, for example, significant localized hiatuses in the build-up, or erosion of the lava pile before the lignite-bearing sediments were deposited. In this context it will be of interest to investigate whether the three younger sediment beds in the western part of the peninsula (see Kristjansson *et al.* 1975; McDougall *et al.* 1984) are accompanied by consistent or variable polarity zone patterns.

As a result of activity in a large volcanic complex south of Dyrafjördur, the rate of build-up of lavas appears to have been particularly high in profiles TF, TH and TI during a normal-polarity chron following the deposition of lignite sediments: the resulting normal-polarity zone in the lava pile may be a partial source of a wide positive aeromagnetic anomaly crossing Arnarfjördur and adjacent fjords (see Jonsson *et al.* 1991, fig. 4). Otherwise, neither the pattern of magnetic polarities nor the distribution of directional groups in Figs 2–4 provide definite indications of, for example, whether the rate of build-up changed systematically at the level of the sediments.

It is not possible to estimate from the palaeomagnetic results the length of the interval represented by the sediments, but those sediments currently in place seem from our palaeomagnetic observations at three sites to have been emplaced in a period shorter than 0.1 Myr at each locality. The polarities and other magnetic properties in the lavas above and below the sediments do not preclude the possibility that the entire interval between emplacement of these lavas was of the order of 0.1–0.3 Myr as suggested by Hardarson *et al.* (1997). More detailed work on radiometric ages, on changes in lava chemistry and alteration, and on changes in tilt at the level of the sediments, is needed before the significance of the sediments in the geological history of the NW peninsula can be assessed.

We have not attempted a correlation here with geomagnetic polarity timescales such as that of Cande & Kent (1995). One reason is that available radiometric dates are not sufficiently accurate or consistent: the age of the lignite sediments can only be estimated as being around 15 Myr with an error of up to 0.5 Myr. Another reason is that in the derivation of these polarity timescales from ocean-ridge magnetic anomalies, the rate of geomagnetic reversals may have been underestimated (see, e.g., Kristjansson & McDougall 1982; Singer & Brown 2002).

Finally, the results on VGP scatter presented in Fig. 7 agree with previous suggestions (e.g. Kristjansson 1995) that the directional stability of the geomagnetic field in Iceland has been increasing over the past 15 Myr.

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